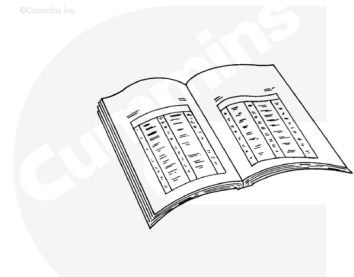


Trainings ▾

Preparatory Steps

⚠ CAUTION ⚠

Use extreme care when releasing the oil pan gasket from the rear cover to prevent damage to the gasket. If the gasket is damaged, the oil pan must be removed and the gasket replaced.



ck800wa

- Disconnect the driven equipment. Refer to the OEM instructions.
- Remove the flywheel. Refer to Procedure 016-005 (Flywheel) in Section 16.
(/qs3/pubsys2/xml/en/procedures/39/39-016-005-tr-gc.html)
- Remove the flywheel housing. Refer to Procedure 016-006 (Flywheel Housing) in Section 16.
(/qs3/pubsys2/xml/en/procedures/39/39-016-006-tr-gc.html)
- Loosen the lubricating oil pan mounting capscrews four revolutions.
- Insert a feeler gauge or shim stock between the rear cover and the oil pan gasket. Move the feeler gauge or the shim stock back and forth to release the gasket from the rear cover.
- Remove the capscrews from the rear cover, and remove the cover from the crankshaft flange.
- Remove the seal from the rear cover.

Remove

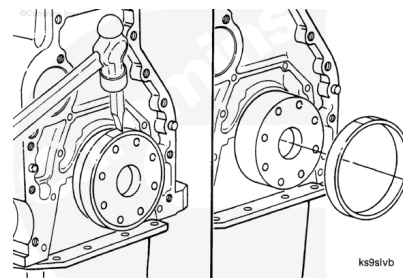
⚠ CAUTION ⚠

Do not nick or gouge the crankshaft with the chisel. If the crankshaft is damaged, it must be replaced.

If the crankshaft presently has a wear sleeve, it **must** be removed before installing a new one.

Use a dull chisel that is **only** as wide as the wear sleeve.

Make one or two soft blows with a hammer to make chisel marks across the wear sleeve. This will expand the wear sleeve, allowing the sleeve to be removed.

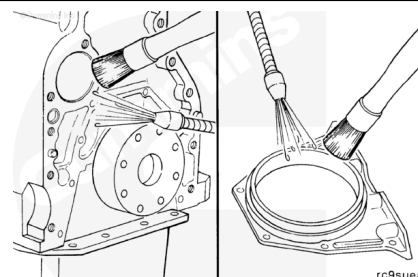


Clean and Inspect for Reuse

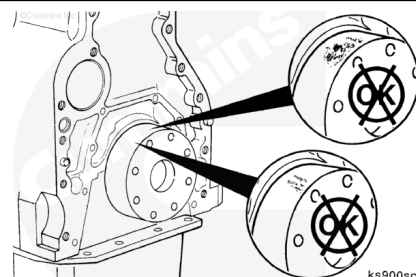
Clean the gasket surface of the cylinder block and rear cover.

Use a crocus cloth to remove any rust or other deposits from the crankshaft flange.

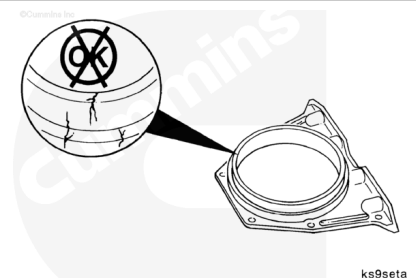
Use a clean cloth to clean the crankshaft flange.



Inspect the crankshaft flange for dirt or nicks.

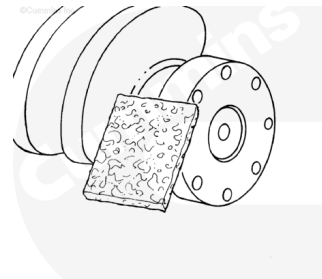


Inspect the rear cover for cracks or other damage.



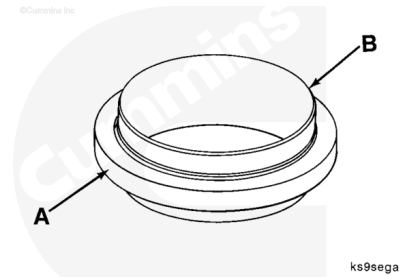
Install

Do **not** use any kind of lubricant to install the seal. The oil seal **must** be installed with the lip of the oil seal and the crankshaft clean and dry to secure proper oil sealing.



ks900eh

The combination crankshaft oil seal (A) wear sleeve (B) replacement kit for service usage is installed on the crankshaft as an assembly. The crankshaft rear oil seal **must not** be removed from the crankshaft rear seal wear sleeve.

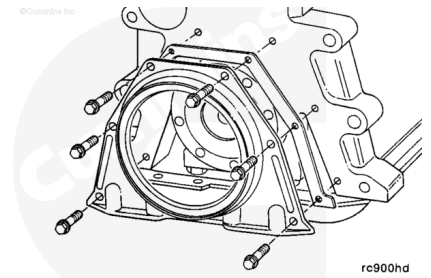


ks9sega

Install the rear cover and gasket.

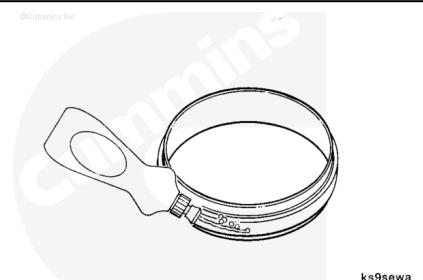
Install the rear cover capscrews. Do **not** tighten. If the oil pan is installed, loosen the oil pan capscrews to allow clearance for rear cover and gasket clearance.

The seal installation is being used to align the rear cover properly. Do **not** push or force the cover in any direction to prevent irregular seal lip position after seal installation.



rc900hd

The oil seal for a wet flywheel housing requires soap on the outside of the seal case. Nothing is required on the outside of the seal case for dry housings.

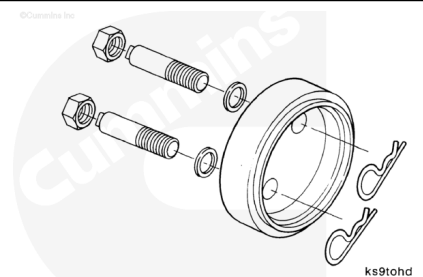


ks9sewa

Use a crankshaft rear seal/wear sleeve installation tool, Part Number 3824078, to install the oil seal/wear sleeve assembly.

Install two (2) threaded studs into the crankshaft capscrew holes.

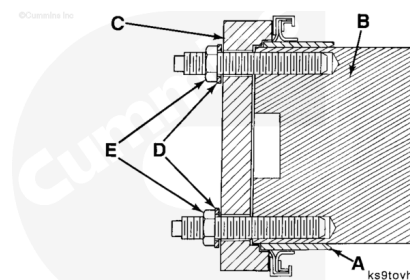
Apply a small amount of clean engine oil to the crankshaft, threaded studs, and inside of the crankshaft rear seal/wear sleeve installation tool.



ks9tohd

Position the chamfered end of the wear sleeve (A) onto the end of the crankshaft (B). Position the counterbore end of the installation tool (C) over threaded studs and align with wear sleeve, perpendicular to the end of the crankshaft. Install the washers (D) and nuts (E) onto the threaded studs.

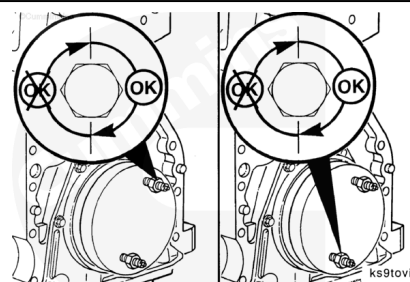
Alternately tighten the nuts until the installation tool contacts the end of the crankshaft.



Do **not** exceed 1/2 revolution of each nut to prevent wear sleeve binding and irregular stretch.

Torque Value: 20 N•m [180 in-lb]

Remove the installation tool and threaded studs.

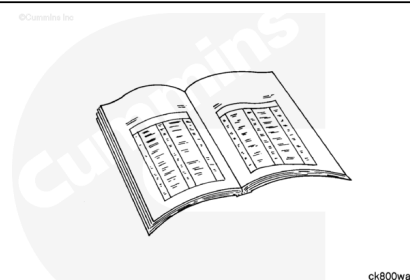


Align the rear cover even with both sides of the oil pan rail on the cylinder block.

The chart below shows the proper torque value when using either a 9.8 or a 10.9 grade bolt. The grade is embossed on the top of each bolt.

Apply Dri Lock to both part numbers below during installation. Dri Lock 205 is recommended. Use Dri Lock 204 or 211, when 205 is **not** available.

Tighten the rear cover capscrews.



Rear Seal Carrier				
Bolt Number	Bolt Size	Bolt Class	Torque	Dri-Lock
3913638	M-6	9.8	10 N•m [in-lb]	No
3991306	M-6	10.9	13 N•m [in-lb]	Yes

Trim the gaskets even with the oil pan mounting surface.

Make sure the gasket trim does **not** enter the engine.

Fill the joint between the lubricating oil pan rail and the rear cover with Three-Bond RTV sealant, Part Number 3823494.

Install the four rear oil pan mounting capscrews to the pan.

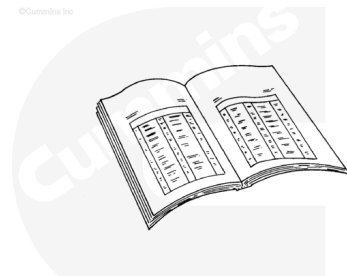
The lubricating oil pan capscrews **must** be tightened within 5 minutes after applying the sealant.

Torque Value: 24 n•m [212 in-lb]

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the flywheel housing. Refer to Procedure 016-006 (Flywheel Housing) in Section 16. (</qs3/pubsys2/xml/en/procedures/39/39-016-006-tr-gc.html>)
- Install the flywheel. Refer to Procedure 016-005 (Flywheel) in Section 16. (</qs3/pubsys2/xml/en/procedures/39/39-016-005-tr-gc.html>)
- Connect the driven equipment. Refer to the OEM instructions.
- Operate the engine until the coolant temperature reaches 82°C [180°F]. Check for leaks and proper operation.